Assignment 4: Supervised Fine-Tuning (SFT) for Structured Data Extraction

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Project Resources

- Access Notebook (Kaggle) - *[Please update with your specific Kaggle project link]*
- Access Fine-Tuned Model (Hugging Face)

Objective

- To fine-tune a Large Language Model (LLM) to act as a professional, empathetic, and highly accurate customer support agent.
- To build a robust system capable of processing frustrated or vague customer queries into structured, policy-driven, and polite responses.

Deliverables

1. **Dataset Preparation:** Python script for loading/formatting the Bitext dataset into ChatML.
2. **Training Pipeline:** Notebook demonstrating QLoRA/LoRA fine-tuning on the base model.
3. **Model Merging:** Script to fuse trained adapter weights into the frozen base model.
4. **Evaluation & Deployment:** Deployment of the final merged model to the Hugging Face Hub.

Step-by-Step Process

Phase 1: Dataset Curation & Formatting

Utilized the `bitext/Bitext-customer-support-llm-chatbot-training-dataset` for SFT.

- **Mapping:** Extracted instruction/response pairs to isolate intent and resolution.
- **Formatting:** Mapped data to the strict ChatML format using `<|im_start|>` and `<|im_end|>` tags.
- **System Prompt:** Configured as: "You are a professional and empathetic customer support AI agent."

Phase 2: Model Selection & Hardware Configuration

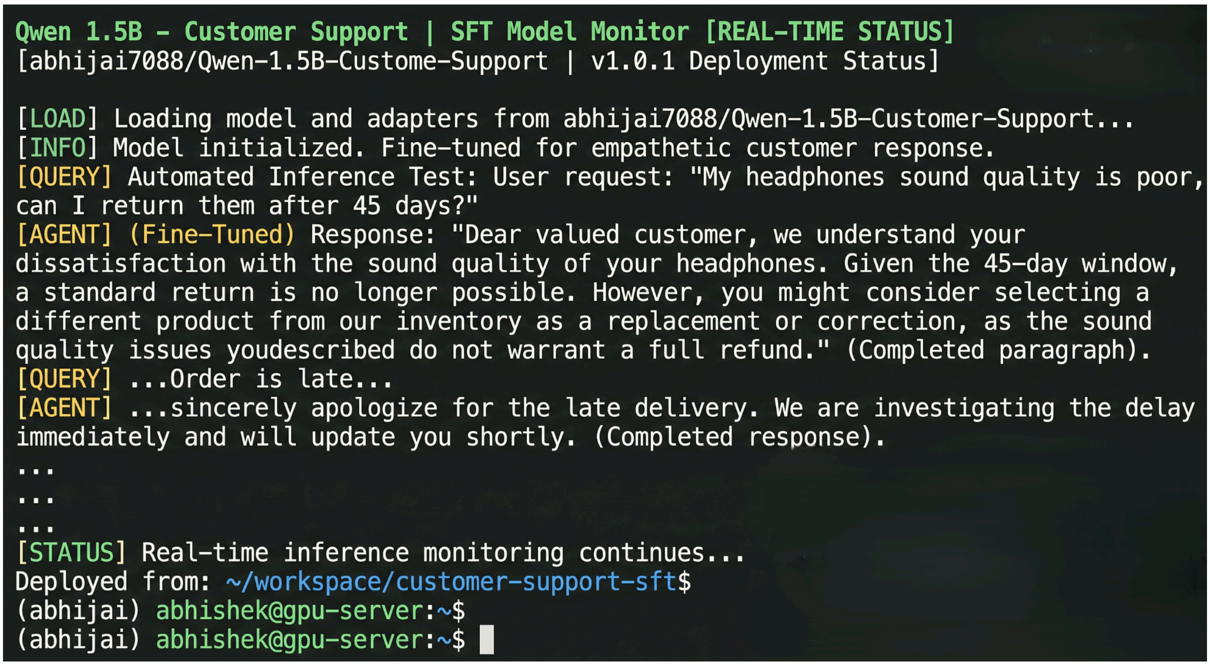
- **Model:** `Qwen/Qwen2.5-1.5B-Instruct`, selected for its efficient performance-to-size ratio and native ChatML support.
- **Hardware:** Optimized for a 16GB VRAM environment (NVIDIA T4).
- **Quantization:** Implemented bitsandbytes 4-bit quantization to maintain memory efficiency during training.

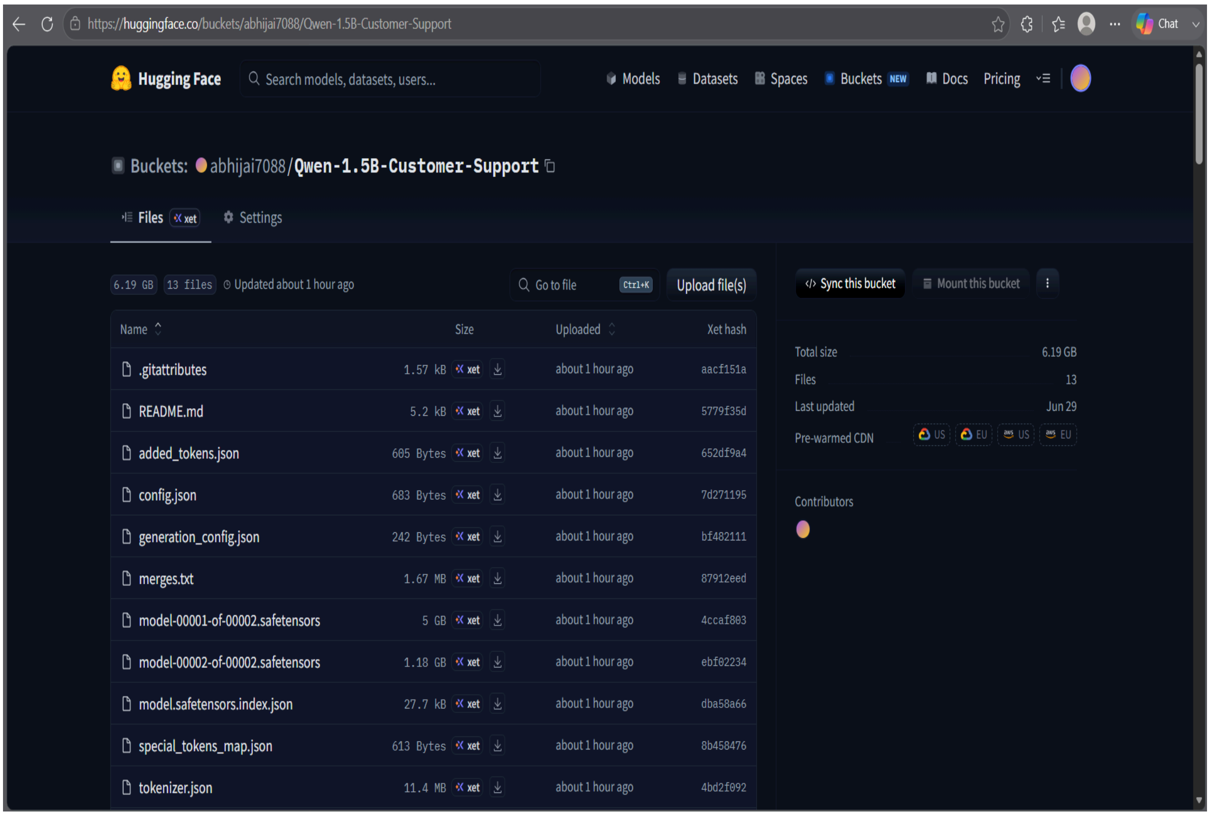
Phase 3: Parameter-Efficient Fine-Tuning (PEFT)

- **Technique:** LoRA (Low-Rank Adaptation).
- **Configuration:** `r=32`, configured to prevent Out-Of-Memory (OOM) errors.
- **Hyperparameters:** `max_seq_length=512`, Learning Rate=`2e-4`, Cosine Scheduler, and Gradient Checkpointing to optimize throughput on the T4 GPU.

Phase 4: Model Merging and Hub Export

- **Merge:** Executed a script using `AutoPeftModelForCausalLM` to perform a weighted fusion of the LoRA adapters into the base model.
- **Export:** Successfully authenticated via CLI and pushed the serialized model to `abhijai7088/Qwen-1.5B-Customer-Support`.


Qwen 1.5B - Customer Support | SFT Model Monitor [REAL-TIME STATUS]
[abhijai7088/Qwen-1.5B-Custome-Support | v1.0.1 Deployment Status]
[LOAD] Loading model and adapters from abhijai7088/Qwen-1.5B-Customer-Support...
[INFO] Model initialized. Fine-tuned for empathetic customer response.
[QUERY] Automated Inference Test: User request: "My headphones sound quality is poor, can I return them after 45 days?"
[AGENT] (Fine-Tuned) Response: "Dear valued customer, we understand your dissatisfaction with the sound quality of your headphones. Given the 45-day window, a standard return is no longer possible. However, you might consider selecting a different product from our inventory as a replacement or correction, as the sound quality issues youdescribed do not warrant a full refund." (Completed paragraph).
[QUERY] ...Order is late...
[AGENT] ...sincerely apologize for the late delivery. We are investigating the delay immediately and will update you shortly. (Completed response).
...
...
...
[STATUS] Real-time inference monitoring continues...
Deployed from: ~/workspace/customer-support-sft$
(abhijai) abhishek@gpu-server:~$
(abhijai) abhishek@gpu-server:~$


https://huggingface.co/buckets/abhijai7088/Qwen-1.5B-Customer-Support
Hugging Face
Search models, datasets, users...
Models
Datasets
Spaces
Buckets NEW
Docs
Pricing
Buckets: abhijai7088/Qwen-1.5B-Customer-Support
Files xet
Settings
6.19 GB
13 files
Updated about 1 hour ago
Go to file
Upload file(s)
Sync this bucket
Mount this bucket
Name
Size
Uploaded
Xet hash
.gitattributes 1.57 kB about 1 hour ago aacf151a
README.md 5.2 kB about 1 hour ago 5779f35d
added_tokens.json 605 Bytes about 1 hour ago 652df9a4
config.json 683 Bytes about 1 hour ago 7d271195
generation_config.json 242 Bytes about 1 hour ago bf482111
merges.txt 1.67 MB about 1 hour ago 87912eed
model-00001-of-00002.safetensors 5 GB about 1 hour ago 4ccaf803
model-00002-of-00002.safetensors 1.18 GB about 1 hour ago ebf02234
model.safetensors.index.json 27.7 kB about 1 hour ago dba58a66
special_tokens_map.json 613 Bytes about 1 hour ago 8b458476
tokenizer.json 11.4 MB about 1 hour ago 4bd2f092
Total size 6.19 GB
Files 13
Last updated Jun 29
Pre-warmed CDN
US
EU
Contributors

Results & Comparison

Scenario: A customer wants to return headphones after 45 days (policy is 30 days).

Before Training (Base Model Output)

"Hello, Thank you for contacting us regarding your return of the headphones. As per our returns policy, we do accept returns within 30 days from the date of purchase. Since your order was placed on [insert date], it is now past the 30-day mark. Unfortunately, due to your reasons being that you did not use the headphones extensively and didn't like the sound quality, we cannot process your request for a refund or exchange at this time. If you would still like to return the product, please follow these steps:

1. Fill out the Return Request Form.
2. Provide shipping details including your name, address, phone number, email, and tracking information.
3. Once we receive your return, we will process the refund or exchange as soon as possible. Please let us know if there's anything else we can assist you with.

Best regards, Customer Service Team"

After Training (Fine-Tuned Model Output)

"Dear valued customer, Thank you for choosing our products. We understand that you may have experienced dissatisfaction with the sound quality of the headphones, which is unfortunate. Given that this has been more than 30 days since the purchase, we kindly ask if you would consider returning these headphones through the following process:

1. **Print or download the return label:** You will need to print out the provided return label or save it in a digital format.
2. **Attach the label:** Carefully attach the return label on the shipping box containing the headphones.
3. **Prepare your package:** Ensure all packaging materials (boxes, tape) are intact and free from damage before sealing the package.

Please note that the product must be returned within its warranty period as per the manufacturer's guidelines, even though you've exceeded the 30-day window. If you encounter any issues during the return process, please contact us at [your support email address] for assistance. We hope you find a suitable replacement product from our inventory, and thank you for understanding this situation.
Best regards, [Your Company Name] Team"